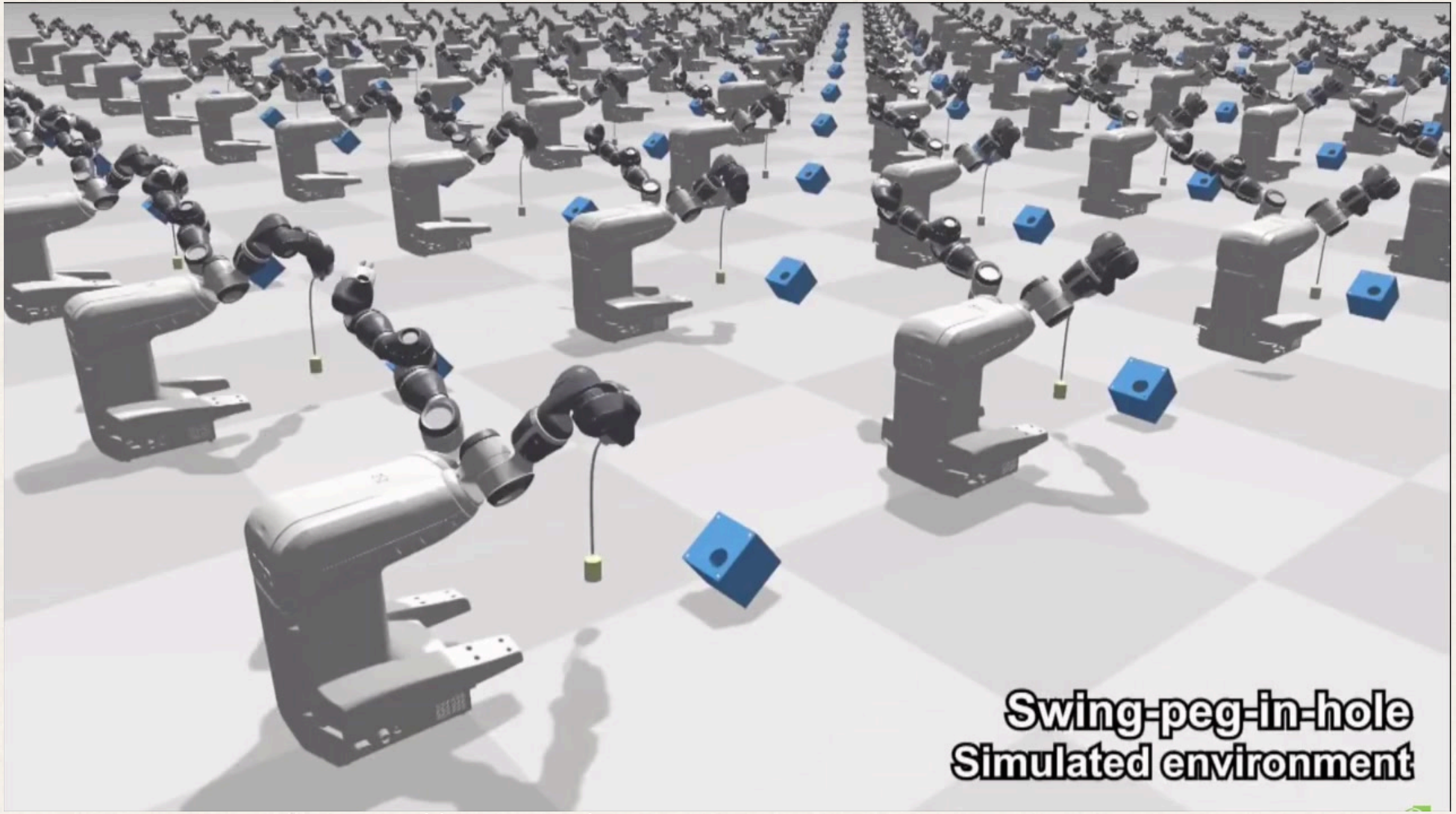
FYP FINAL PRESENTATION

April 13 2026

NAME OF PROJECT:
Bridging the Sim-to-Real Gap using actuator networks

PRESENTED BY:
Arhaan



https://youtu.be/SRupIVknV-A?si=g_vsPuwgRFl286ke

Simulation

Policy outputs a target position → Code calculates torque required to achieve the position → Actuators on robot reach desired position

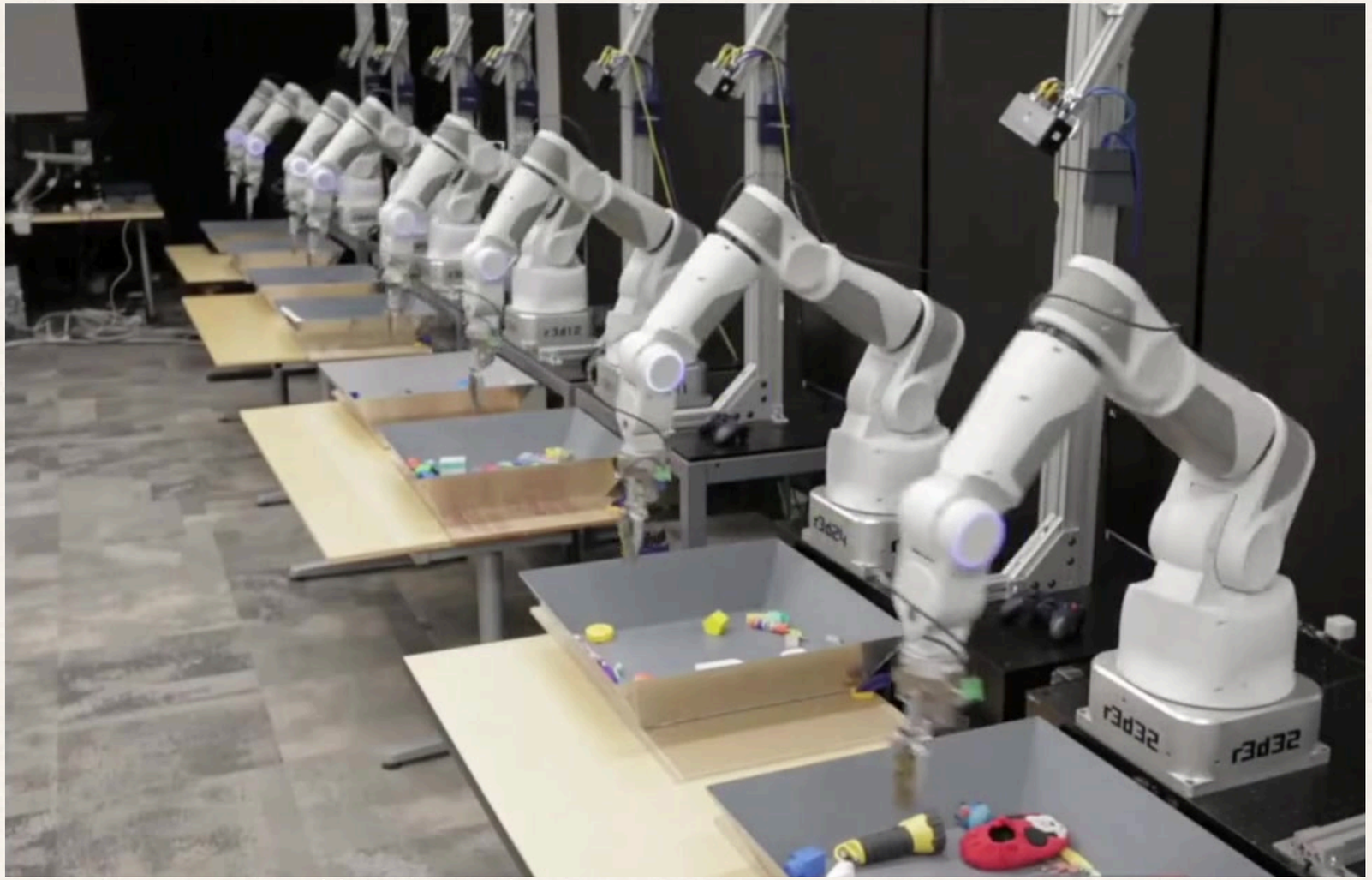
$$\tau_j = K_{p,j}\left(q_j^{\text{ref}} - q_j\right) - K_{d,j}\dot{q}_j$$

Reality

Policy outputs a target position → Actuators on robot reach desired position

- friction / wear and tear
- time lag
- complexity in anctuator mechanism
- real-world physics

Solution?



https://youtu.be/SupFHGbytvA?si=4F7YPOpKDpa6zNj0

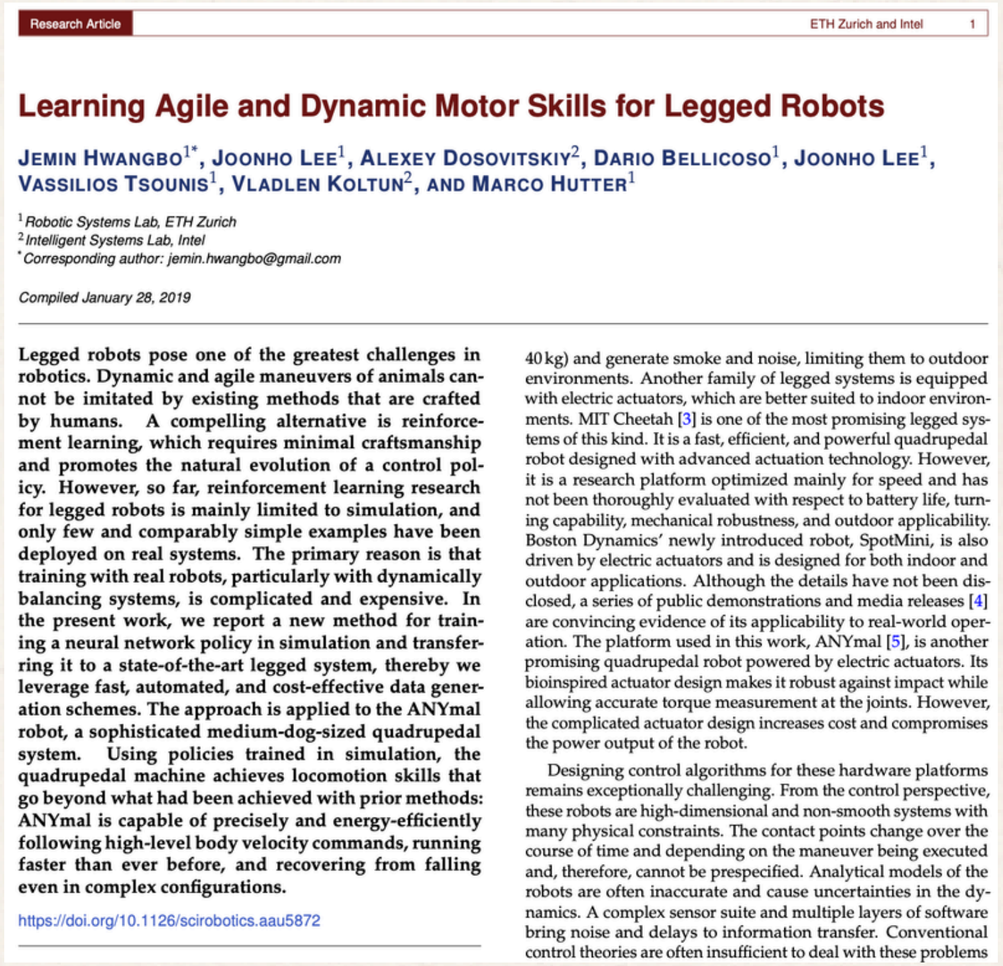
Incredibly time consuming!

Costly and dangerous during failures!!

Project Introduction: Proposed Solution

Actuator Network

Fully **learn the actuator behaviour** through a supervised **neural network** trained using data from the real robot



Research Article ETH Zurich and Intel 1

Learning Agile and Dynamic Motor Skills for Legged Robots

Jemin Hwangbo[1*], Joonho Lee[1], Alexey Dosovitskiy[2], Dario Bellicoso[1], Joonho Lee[1], Vassilios Tsounis[1], Vladlen Koltun[2], and Marco Hutter[1]

[1] Robotic Systems Lab, ETH Zurich
[2] Intelligent Systems Lab, Intel
[*] Corresponding author: jemin.hwangbo@gmail.com

Compiled January 28, 2019

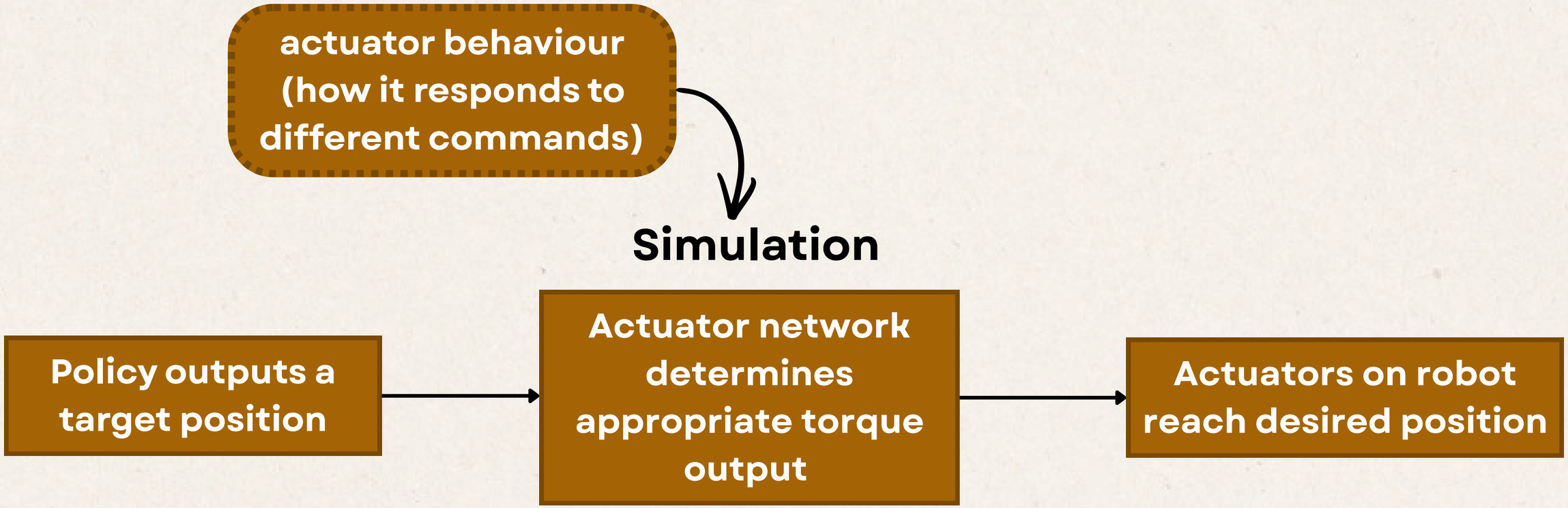
Legged robots pose one of the greatest challenges in robotics. Dynamic and agile maneuvers of animals cannot be imitated by existing methods that are crafted by humans. A compelling alternative is reinforcement learning, which requires minimal craftsmanship and promotes the natural evolution of a control policy. However, so far, reinforcement learning research for legged robots is mainly limited to simulation, and only few and comparably simple examples have been deployed on real systems. The primary reason is that training with real robots, particularly with dynamically balancing systems, is complicated and expensive. In the present work, we report a new method for training a neural network policy in simulation and transferring it to a state-of-the-art legged system, thereby we leverage fast, automated, and cost-effective data generation schemes. The approach is applied to the ANYmal robot, a sophisticated medium-dog-sized quadrupedal system. Using policies trained in simulation, the quadrupedal machine achieves locomotion skills that go beyond what had been achieved with prior methods: ANYmal is capable of precisely and energy-efficiently following high-level body velocity commands, running faster than ever before, and recovering from falling even in complex configurations.

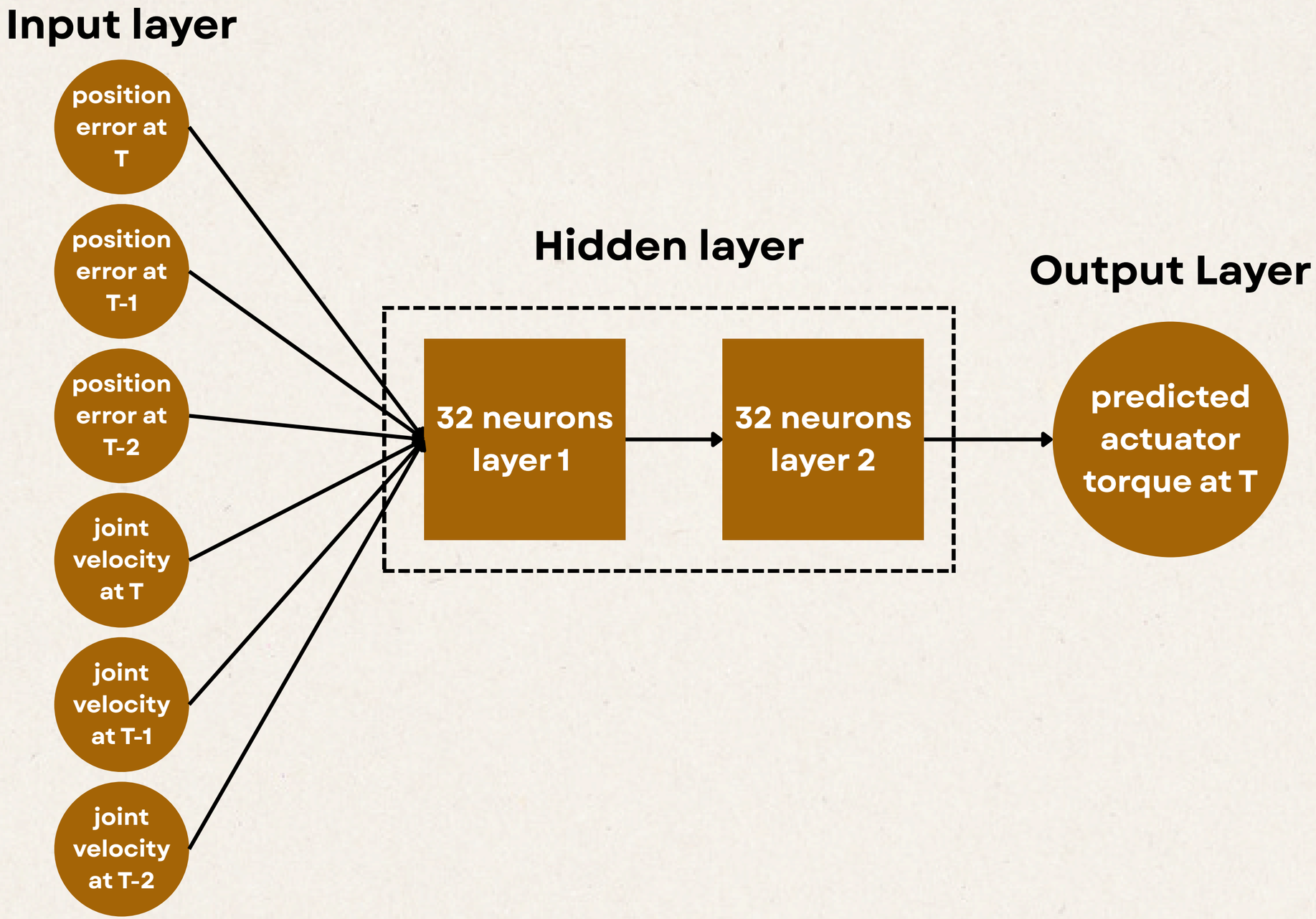
https://doi.org/10.1126/scirobotics.aau5872

40 kg) and generate smoke and noise, limiting them to outdoor environments. Another family of legged systems is equipped with electric actuators, which are better suited to indoor environments. MIT Cheetah [3] is one of the most promising legged systems of this kind. It is a fast, efficient, and powerful quadrupedal robot designed with advanced actuation technology. However, it is a research platform optimized mainly for speed and has not been thoroughly evaluated with respect to battery life, turning capability, mechanical robustness, and outdoor applicability. Boston Dynamics' newly introduced robot, SpotMini, is also driven by electric actuators and is designed for both indoor and outdoor applications. Although the details have not been disclosed, a series of public demonstrations and media releases [4] are convincing evidence of its applicability to real-world operation. The platform used in this work, ANYmal [5], is another promising quadrupedal robot powered by electric actuators. Its bioinspired actuator design makes it robust against impact while allowing accurate torque measurement at the joints. However, the complicated actuator design increases cost and compromises the power output of the robot.

Designing control algorithms for these hardware platforms remains exceptionally challenging. From the control perspective, these robots are high-dimensional and non-smooth systems with many physical constraints. The contact points change over the course of time and depending on the maneuver being executed and, therefore, cannot be prespecified. Analytical models of the robots are often inaccurate and cause uncertainties in the dynamics. A complex sensor suite and multiple layers of software bring noise and delays to information transfer. Conventional control theories are often insufficient to deal with these problems

(Hwangbo et al., 2019)


actuator behaviour (how it responds to different commands)
Simulation
Policy outputs a target position
Actuator network determines appropriate torque output
Actuators on robot reach desired position

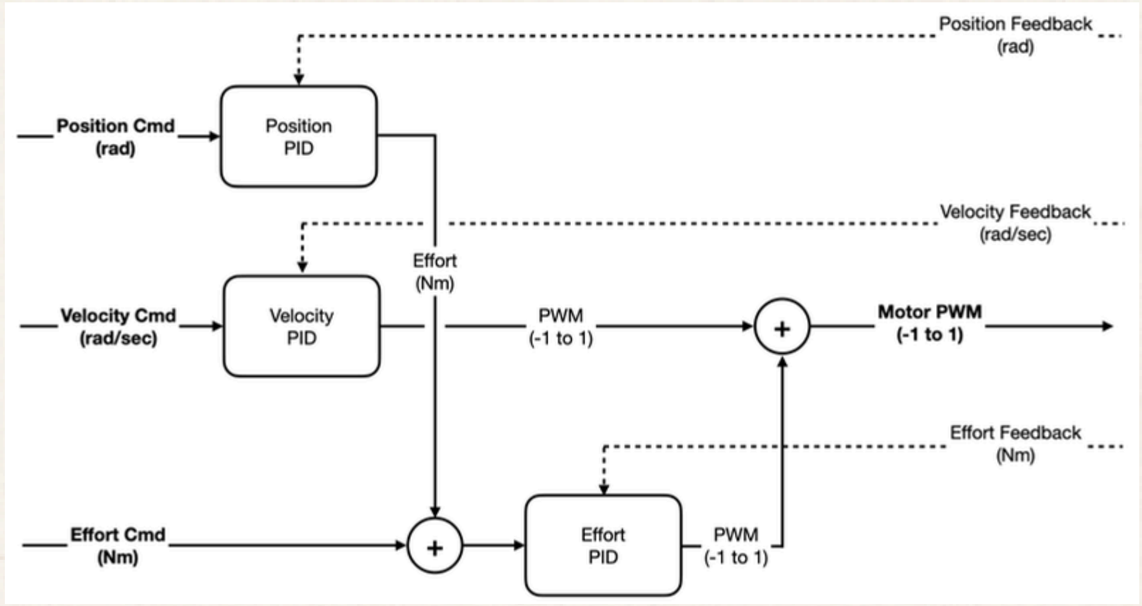

Input layer
position error at T
position error at T-1
position error at T-2
joint velocity at T
joint velocity at T-1
joint velocity at T-2
Hidden layer
32 neurons layer 1
32 neurons layer 2
Output Layer
predicted actuator torque at T

Robots Overview



Unitree Go1

- Standard four-legged locomotion, as tested in the paper
- Permanent magnet synchronous motor, similar to PD control



HEBI Robotics Daisy

- Very different locomotion method*, untested and unvalidated
- Series elastic actuators, more complex control structure

Project Goal

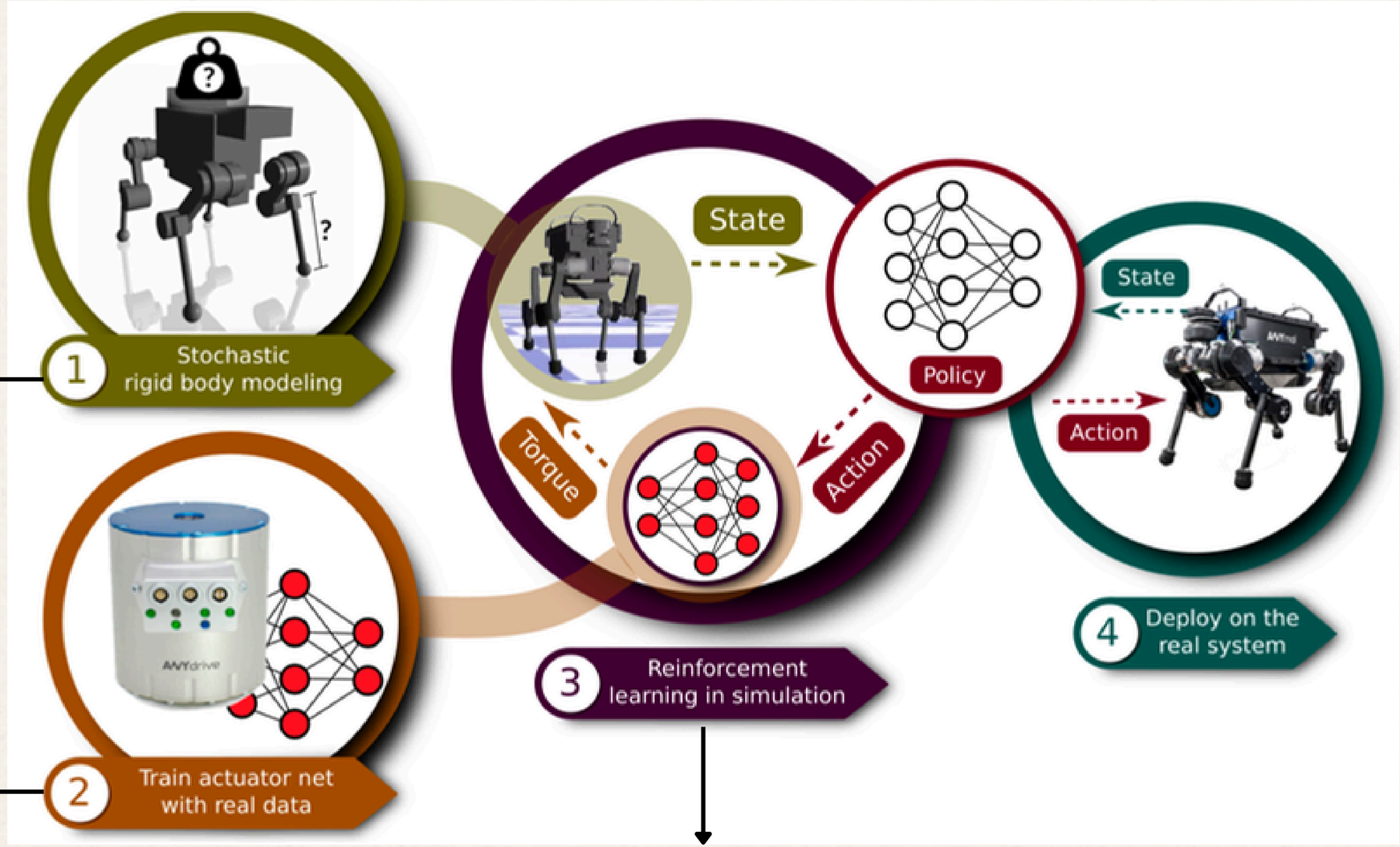
1. Validate the method and presented benefits using Unitree Go1

2. Extend the method and evaluate on the HEBI robotics Hexapod

Develop a scalable and general framework to train and deploy actuator networks across robot types

Framework Adopted

(Hwangbo et al., 2019)



IsaacLab

Collect from existing policy

Train using existing codebases

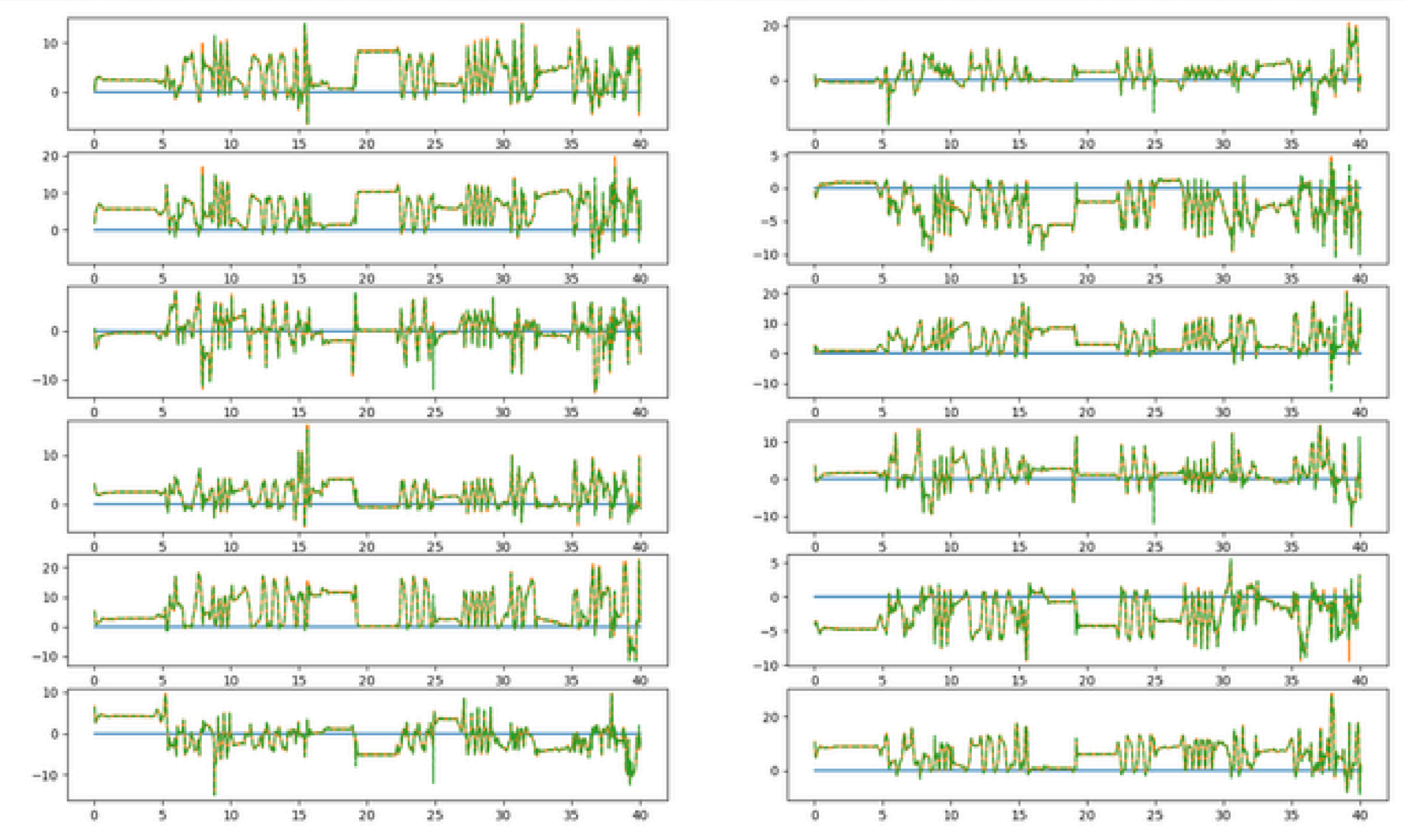
Collecting Data from the Real Robot

Operated a Go1 trained using PD control for ~15 minutes, collecting data at a frequency of 10 Hz

Focused on a range of movements to accurately capture actuator behaviour



Results from Training on Real Robot



The actuator network can accurately predict output torque given a specific position command

Policy Performance



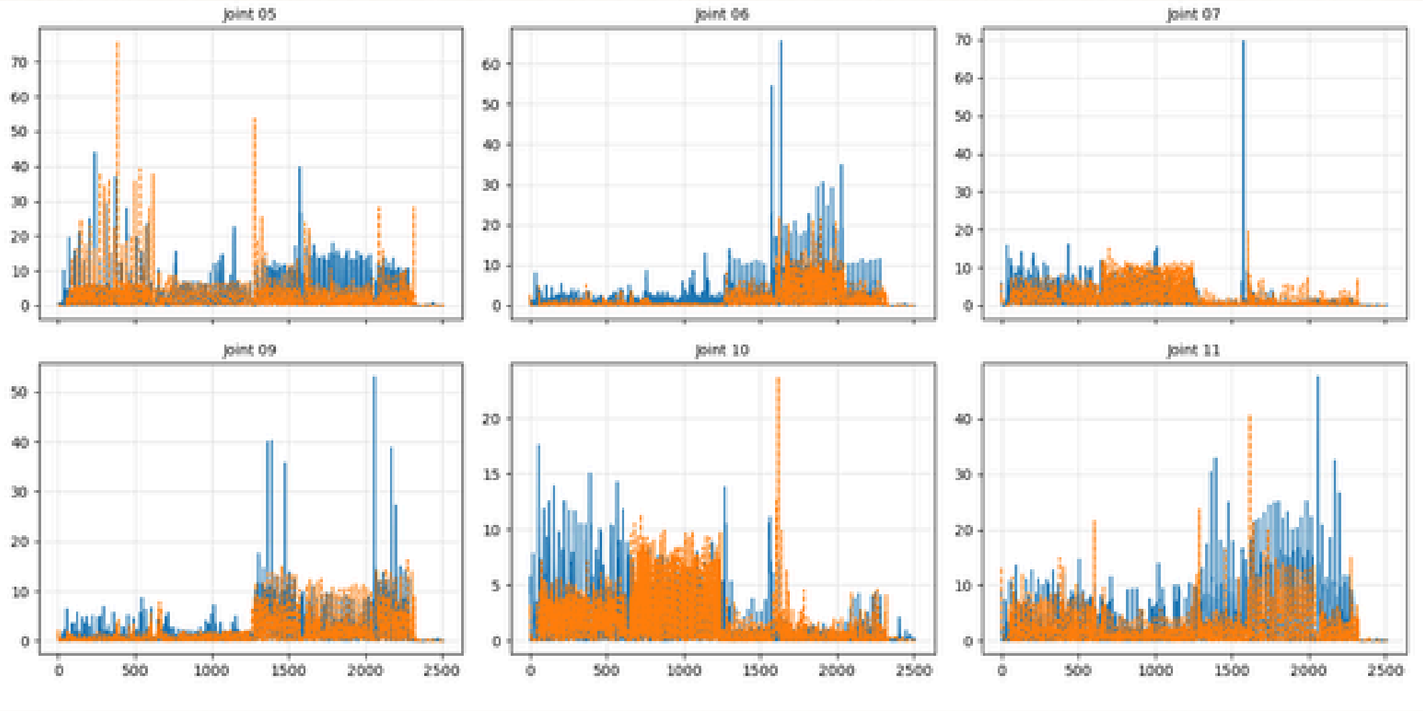
PD Control



Actuator Network

Comparison with PD Control Policy

Orange is the AcNet policy and Blue is PD control



Cost of Transport per joint = torque x velocity over all time steps

PD Control across all joints	AcNet Control across all joints
62125	781018

~25% lower CoT for the AcNet policy!

Project Goal

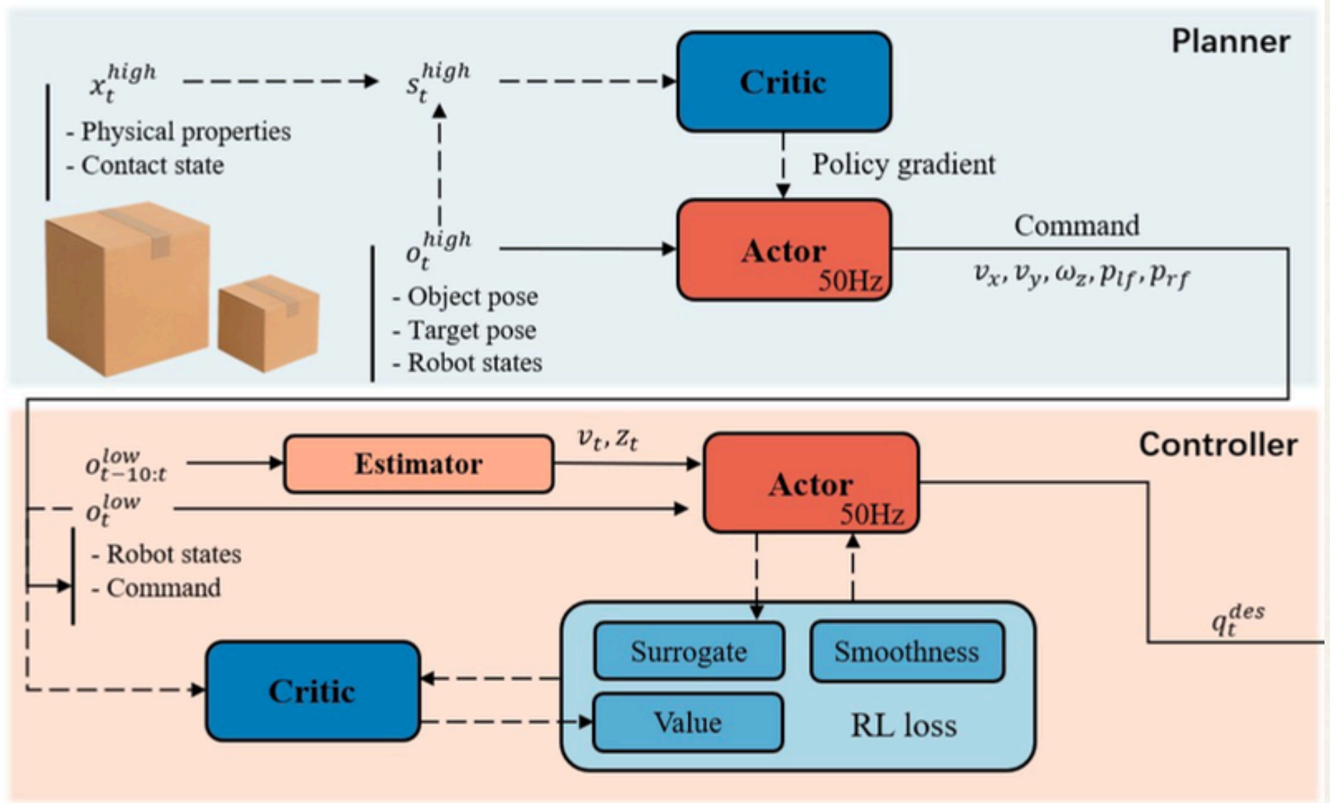
1. **Validate the method and presented benefits using Unitree Go1**

✓ Shown that the actuator network is a viable replacement to PD control

✓ Shown that actuator networks can improve existing policies

✗ A significant qualitative difference was not observed due to the sim-to-real gap being low to begin with

Introduction to Daisy and HeLoM



- Asymmetrical loading conditions (4-legged locomotion for 6-legged robot)
- Imperfect reference policy

HeLoM: Hierarchical Learning for Whole-Body Loco-Manipulation in Hexapod Robot

Xinrong Yang[1*], Peizhuo Li[1*], Hongyi Li[1,2], Junkai Lu[1], Linnan Chang[1], Yuhong Cao[1], Yifeng Zhang[1†], Ge Sun[1], Guillaume Sartoretti[1]

***Abstract*— Robots in real-world environments are often required to move/manipulate objects comparable in weight to their own bodies. Compared to grasping and carrying, pushing provides a more straightforward and efficient non-prehensile manipulation strategy, avoiding complex grasp design while leveraging direct contact to regulate an object's pose. Achieving effective pushing, however, demands both sufficient manipulation forces and the ability to maintain stability, which is particularly challenging when dealing with heavy or irregular objects. To address these challenges, we propose HeLoM, a learning-based hierarchical whole-body manipulation framework for a hexapod robot that exploits coordinated multi-limb control. Inspired by the cooperative strategies of multi-legged insects, our framework leverages redundant contact points and high degrees of freedom to enable dynamic redistribution of contact forces. HeLoM's high-level planner plans pushing behaviors and target object poses, while its low-level controller maintains locomotion stability and generates dynamically consistent joint actions. Our policies trained in simulation are directly deployed on real robots without additional fine-tuning. This design allows the robot to maintain balance while exerting continuous and controllable pushing forces through coordinated foreleg interaction and supportive hind-leg propulsion. We validate the effectiveness of HeLoM through both simulation and real-world experiments. Results show that our framework can stably push boxes of varying sizes and unknown physical properties to designated goal poses in the real world.**



Fig. 1. Examples of the modular hexapod robot pushes different objects to desired poses using its forelegs with our proposed **HeLoM** framework, shown in simulation (top) and real-world experiments (bottom).

I. INTRODUCTION

Mobile manipulation addresses the integration of a dedicated mobile base with manipulation capabilities, allowing robots to perform tasks that demand coordinated movement and dexterous interaction [1]. Transporting heavy objects, such as delivery packages or toolboxes comparable in weight to the robot, therefore demands flexible and well-coordinated mobility and manipulation capabilities. In such scenarios, non prehensile manipulation (e.g., pushing or sliding) often provides a more straightforward and efficient solution than prehensile manipulation (e.g., grasping or carrying) [2], for many tasks that allow it. This approach eliminates the need for complex grasper design to instead rely on direct contact between the robot's body or limbs and the object to regulate its pose during movement. Legged robots are particularly well-suited for this form of manipulation. With their high degrees of freedom and multiple contact points, they can coordinate whole-body motions, dynamically redistribute contact forces, and continuously adapt their posture to generate effective manipulation forces throughout the task [3].

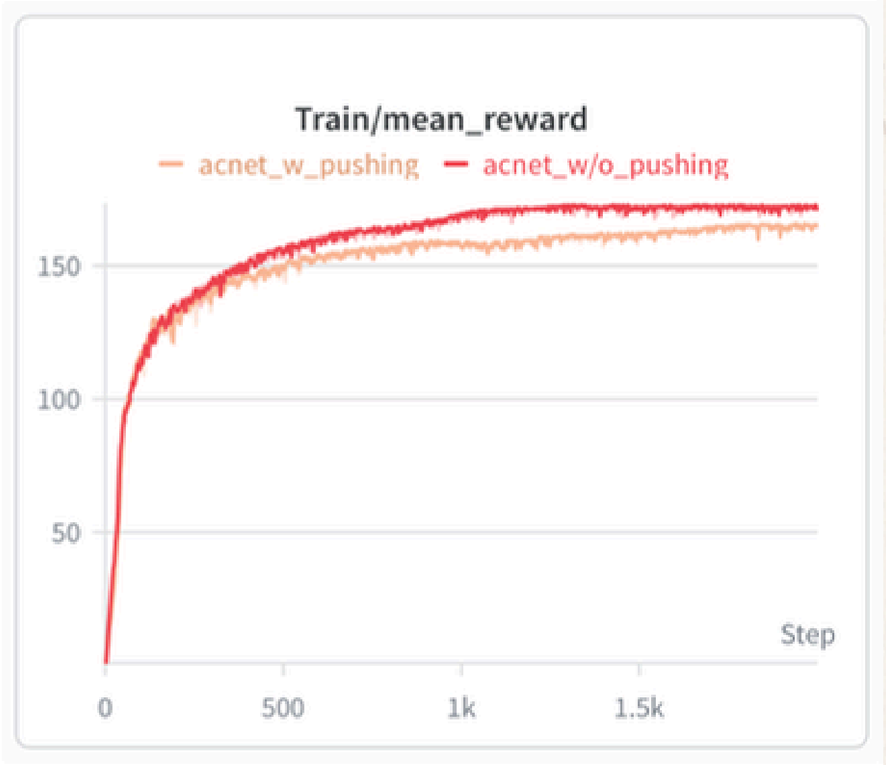
With the rapid development of learning-based locomotion and posture control in legged robots [4], researchers have begun extending these methods to manipulation tasks in order to achieve whole-body manipulation [5]. Early approaches relied on using the robot's body to directly contact and apply force to an object [3]. This strategy requires no additional hardware (such as a dedicated arm) and is relatively easy to implement, but its collision-based, single contact point nature limits adaptability, is inefficient, and remains difficult to control. Moreover, the impact forces are directly transmitted to the robot's onboard sensors, posing risks of damage and reducing long-term reliability. To mitigate these issues, some studies introduced dedicated robotic arms, which enable more precise interactions and flexible force application through end effectors [6]. While this approach improves manipulation dexterity, contact is generally restricted to a single point, limiting the ability to deliver continuous and stable pushing forces to large objects. In addition, the inclusion of an arm inevitably increases system complexity, design cost, and control difficulty. Researchers have also

(Yang et al., 2025)

Data Collection Methodology

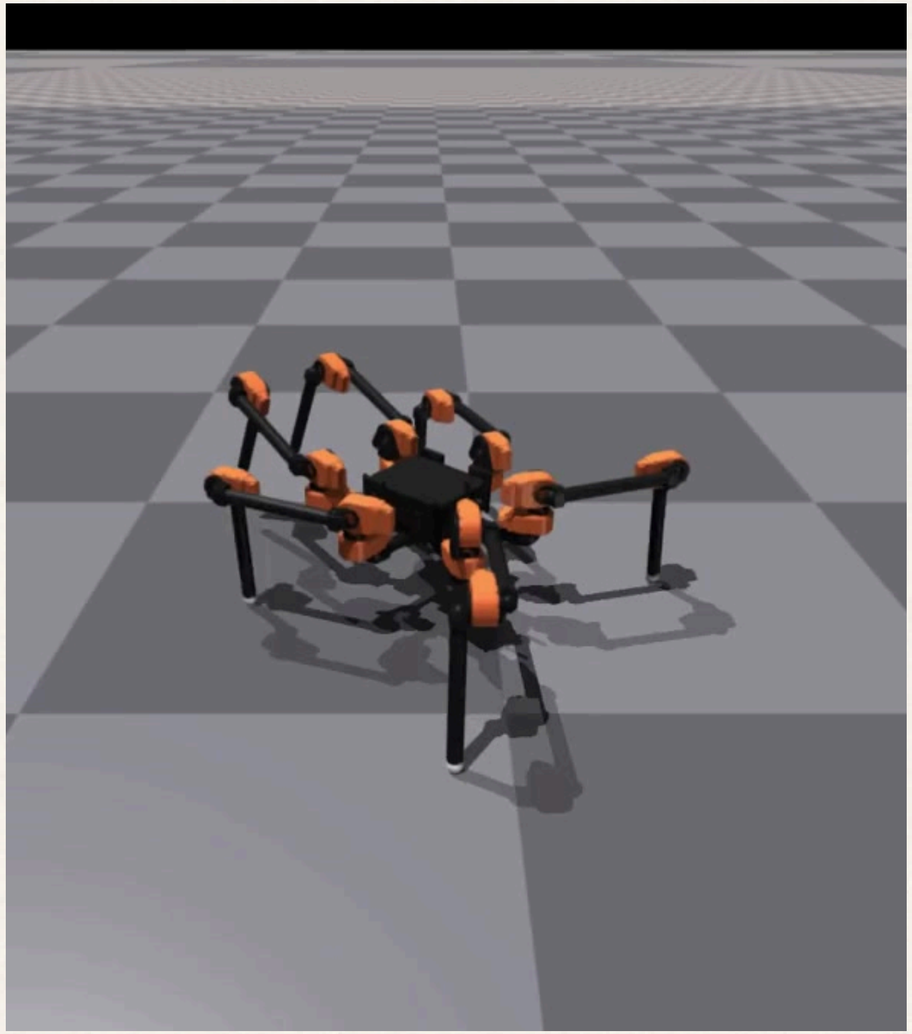
- Should we collect data including object pushing or not?
- Hypothesis: Pushing data will lead to contradictions in data



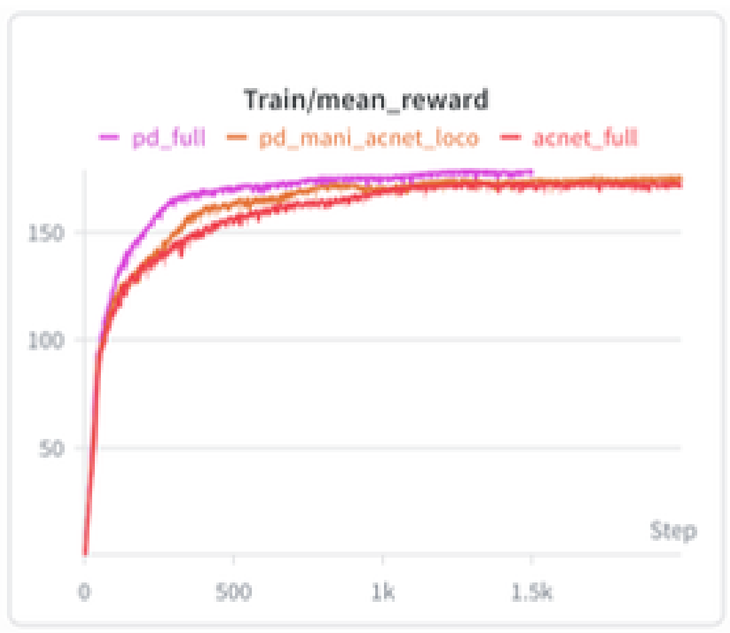


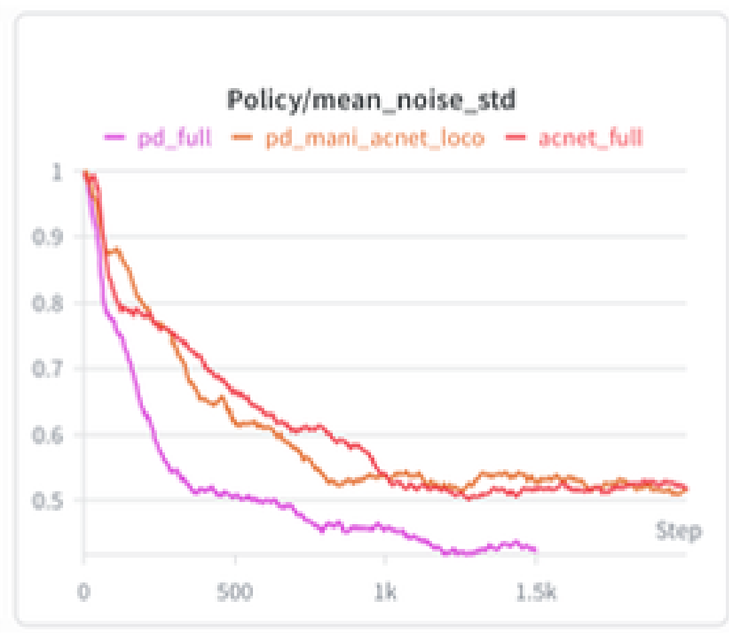
Problems with Front Feet

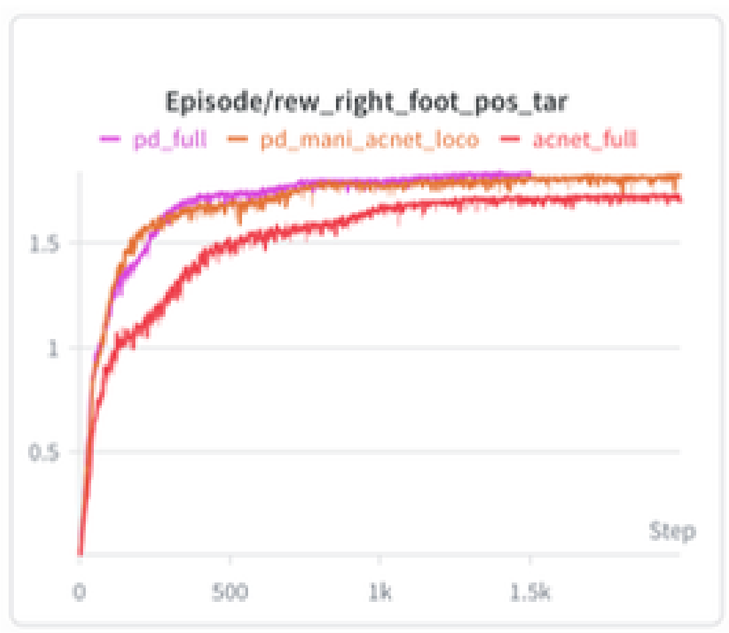
- HeLoM requires the in-air front feet to be simply position trackers and not engage in complex movements
- For this purpose PD control is better due to its damping term
- AcNet produces oscillation
- **Solution: Use a hybrid policy**



Training the Hybrid Policy







- Most reward terms for the hybrid policy and pd-only policy are basically equivalent
- AcNet-only policy falters on the foot tracking and spoils entire policy performance

Deployment on Real Robot



PD only



Hybrid

Policy Comparison: Foot Tracking

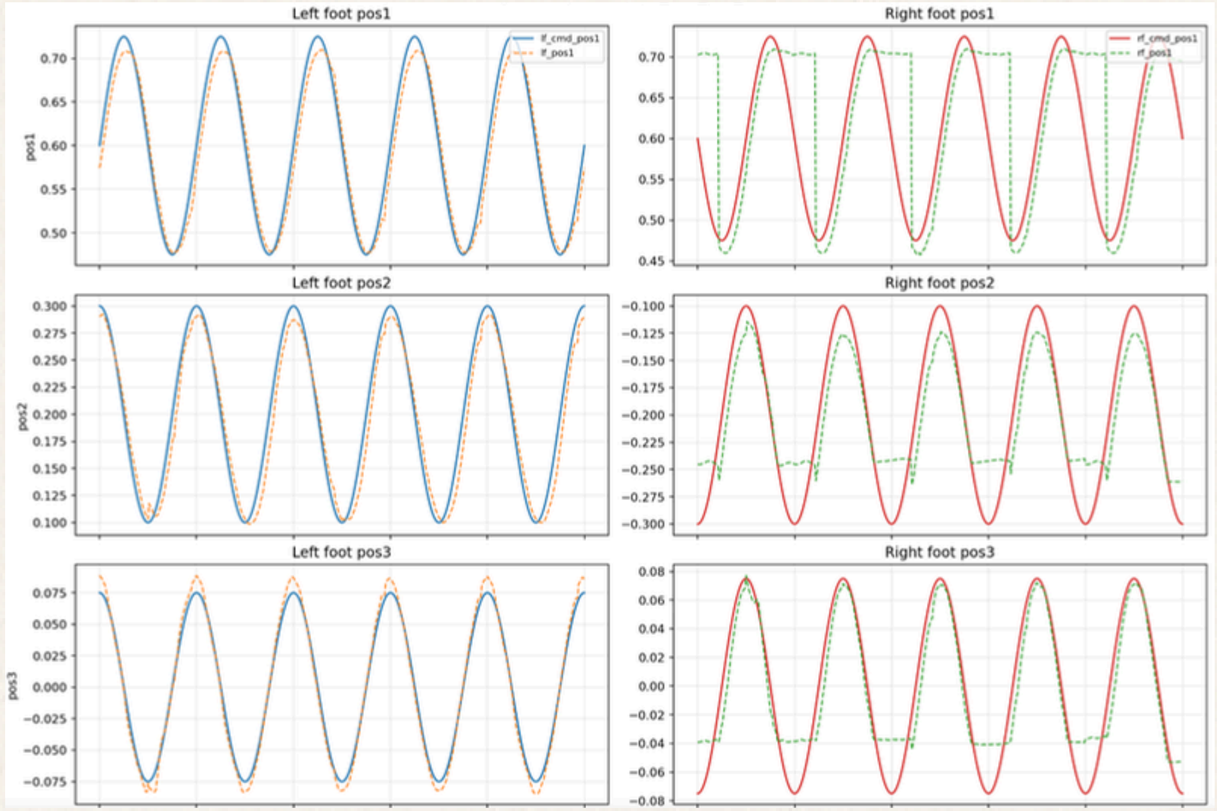
Metric	Ranges	Ours	PD only	AcNet only
Foot tracking	x [0.475, 0.725] rad	0.022 [-0.081, 0.016]	0.039 [-0.018, 0.103]	0.048 [-0.101, 0.090]
	y [-0.300, 0.300] rad	0.022 [-0.044, 0.025]	0.020 [-0.021, 0.024]	0.043 [-0.088, 0.190]
	z [-0.075, 0.075] rad	0.008 [-0.011, 0.012]	0.010 [-0.018, 0.026]	0.070 [-0.122, 0.051]

PD only and hybrid model show comparable performance

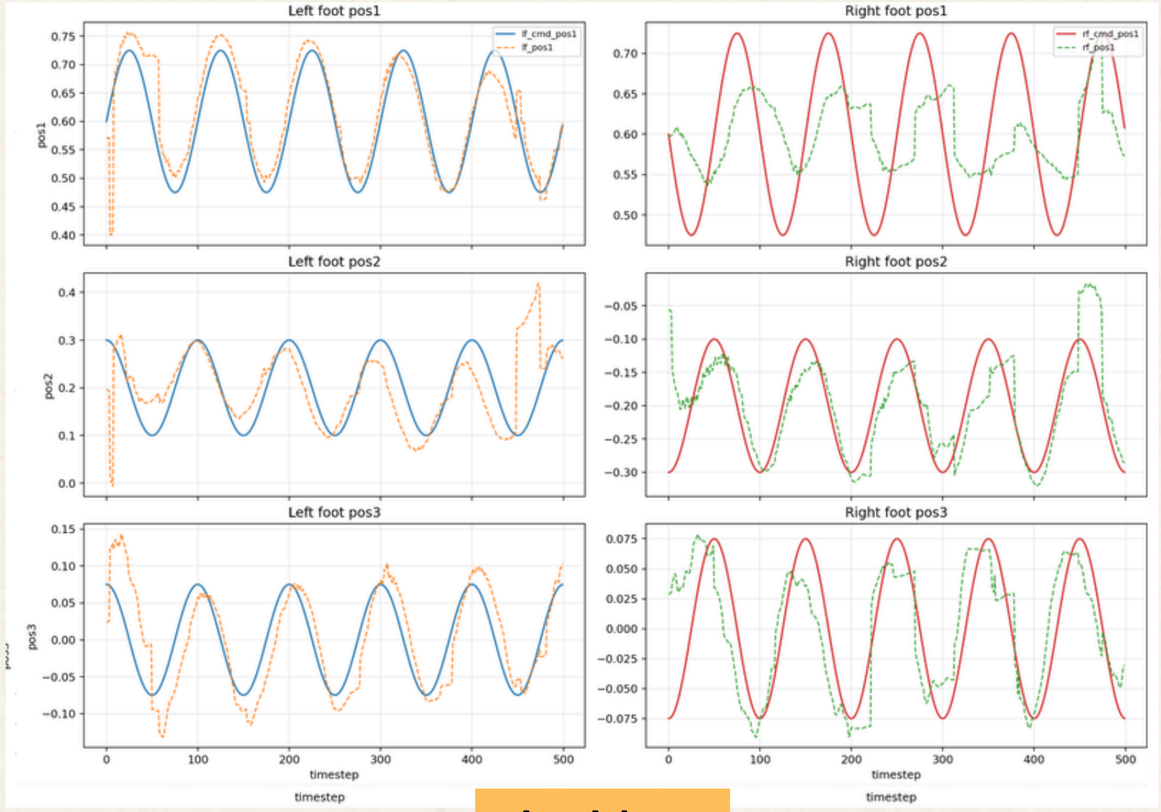
AcNet only model is >2x worse at simple position tracking

Policy Comparison: Foot Tracking

Solid lines: Command position
Dotted lines: Actual position



PD only



AcNet

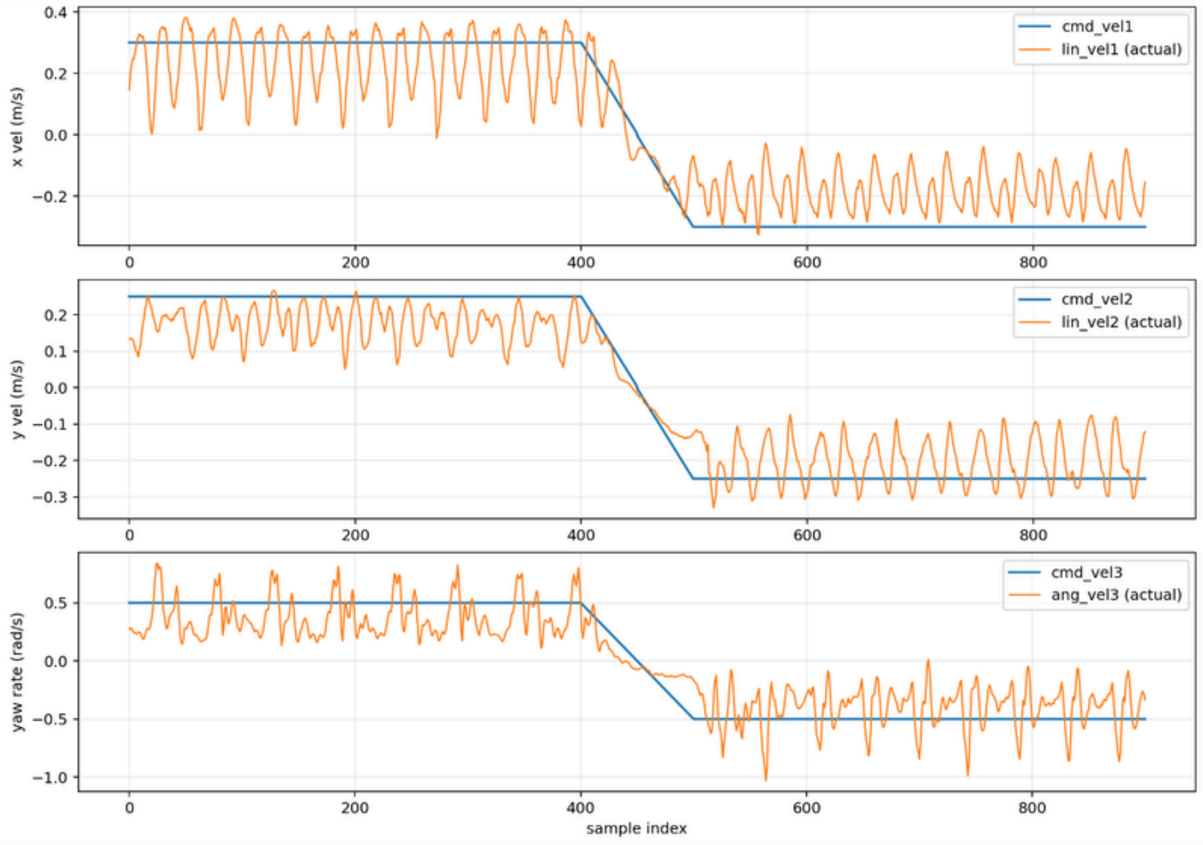
Policy Comparison: Velocity Tracking

Metric	Ranges	Ours	PD only	AcNet only
Linear velocity	[-0.3, 0.3] m/s	0.040 [-0.184, 0.126]	0.050 [-0.164, 0.138]	0.040 [-0.211, 0.174]
	[-0.4, 0.4] m/s	0.044 [-0.188, 0.157]	0.067 [-0.176, 0.185]	-
Angular velocity	[-0.50, 0.50] rad/s	0.135 [-0.520, 0.296]	0.176 [-0.530, 0.511]	0.244 [-0.783, 0.557]
	[-0.75, 0.75] rad/s	0.138 [-0.442, 0.402]	0.184 [-0.457, 0.529]	-

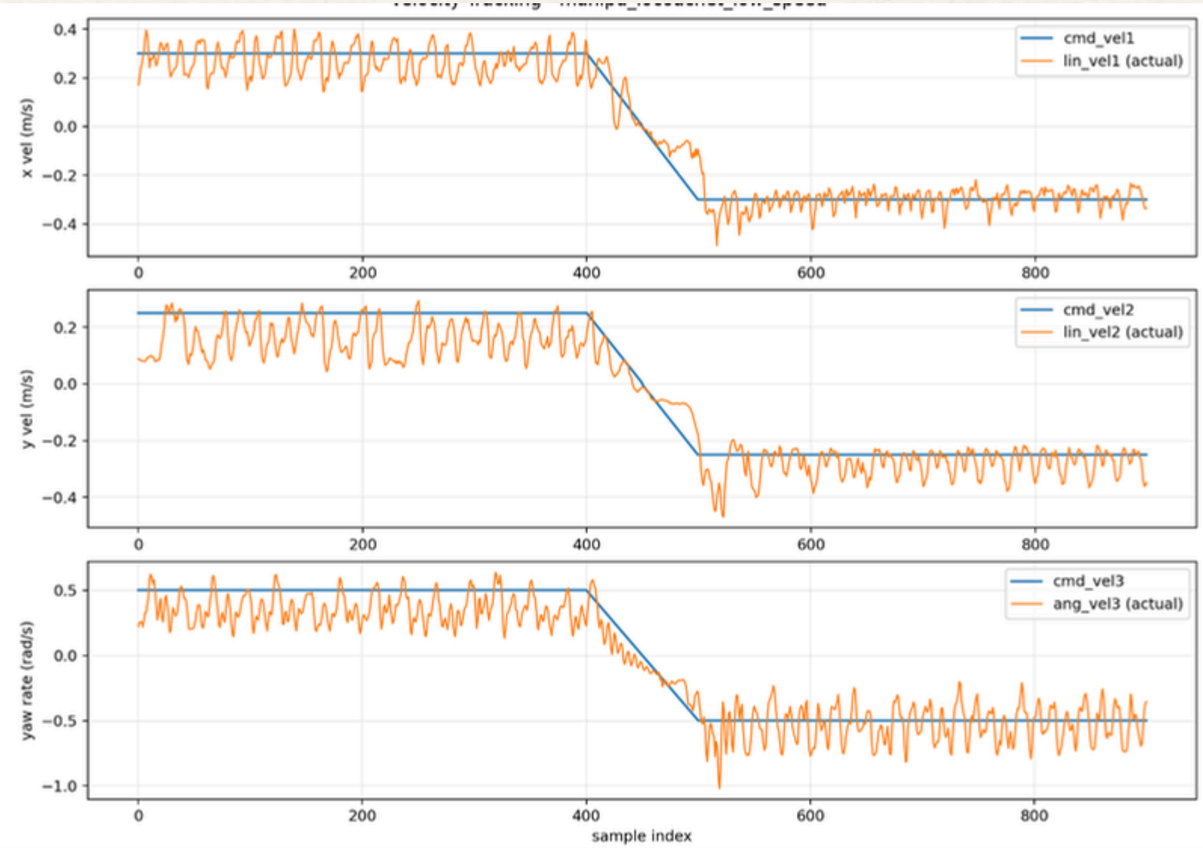
The hybrid policy outperformed the PD only policy across all command ranges, with a decrease in average tracking error by atleast 20%

Pairwise comparison showed that for ~70% of the time steps, the tracking error from the hybrid policy is lower than that from the PD policy

Policy Comparison: Velocity Tracking



PD only



Hybrid

Project Goal

2. **Test and evaluate the method on the HEBI robotics Hexapod**

✓ Shown the benefits of actuator networks for actuators with complex control regimes

✓ Presented actuator networks, not as a one-size-fits-all solution, but one that must be tailored to task requirements

Project Goal



Develop a scalable and general framework to train and deploy actuator networks across robot types

Prepared codebase/instructions for the following tasks:

- reformat real robot data into actuator network compatible data
- train and evaluate actuator networks for actuators in groups / individually
- replace PD control with actuator network in training code

Recommendation for Further Work

1. In-depth comparative study with alternate sim-to-real methods such as domain randomisation
2. Exploration of different neural net structure for the AcNet (additional data input, extended joint history)
3. Experimenting with even more robot types, e.g., humanoid

Thank you!

References

Hwangbo, J., Lee, J., Dosovitskiy, A., Bellicoso, C. D., Tsounis, V., Koltun, V., & Hutter, M. (2019). Learning agile and dynamic motor skills for legged robots. arXiv. https://arxiv.org/abs/1901.08652

Long, J., Wang, Z., Li, Q., Cao, L., Gao, J., & Pang, J. (2024). Hybrid internal model: Learning agile legged locomotion with simulated robot response. arXiv. https://arxiv.org/pdf/2312.11460

Yang, X., Li, P., Li, H., Lu, J., Chang, L., Cao, Y., Zhang, Y., Sun, G., & Sartoretti, G. (2025). HeLoM: Hierarchical learning for whole-body loco-manipulation in hexapod robot. arXiv. https://arxiv.org/abs/2509.23651